



## **PROJECT REPORT**

**Course Name: Applied Model Based Design**

**Course Code: 26ELC332**

### **Model Based Design of Bidirectional Vehicle-to-Grid (V2G) Power Conversion System**

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**CB.EN.U4ELC23019**

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## 1. Abstract & Objectives

**Abstract** This project focuses on the design, simulation, and validation of a bidirectional Vehicle-to-Grid (V2G) power conversion system utilizing Model-Based Design (MBD) methodologies in MATLAB/Simulink and Simscape. The core objective is to develop a smart power interface capable of autonomously switching between Grid-to-Vehicle (G2V) charging and Vehicle-to-Grid (V2G) discharging modes.

By continuously monitoring the real-time State of Charge (SOC) of the Electric Vehicle (EV) battery and responding to external grid demand signals, this system enables the EV to function as a distributed energy storage unit, contributing to grid stabilization.

### Primary Objectives

- Achieve stable, bidirectional power flow between an EV battery and a 3-phase grid infrastructure.
- Implement an autonomous Stateflow controller to handle mode switching (G2V, V2G, and Standby) dynamically based on SOC thresholds and grid demand pulses.
- Maintain battery health by keeping SOC within strict operational boundaries.
- Ensure clean, regulated sinusoidal AC waveforms at the grid interface during both active modes.
- Validate the model configuration, code generation readiness, and execution coverage using Model Advisor, Embedded Coder, and Test Manager frameworks.

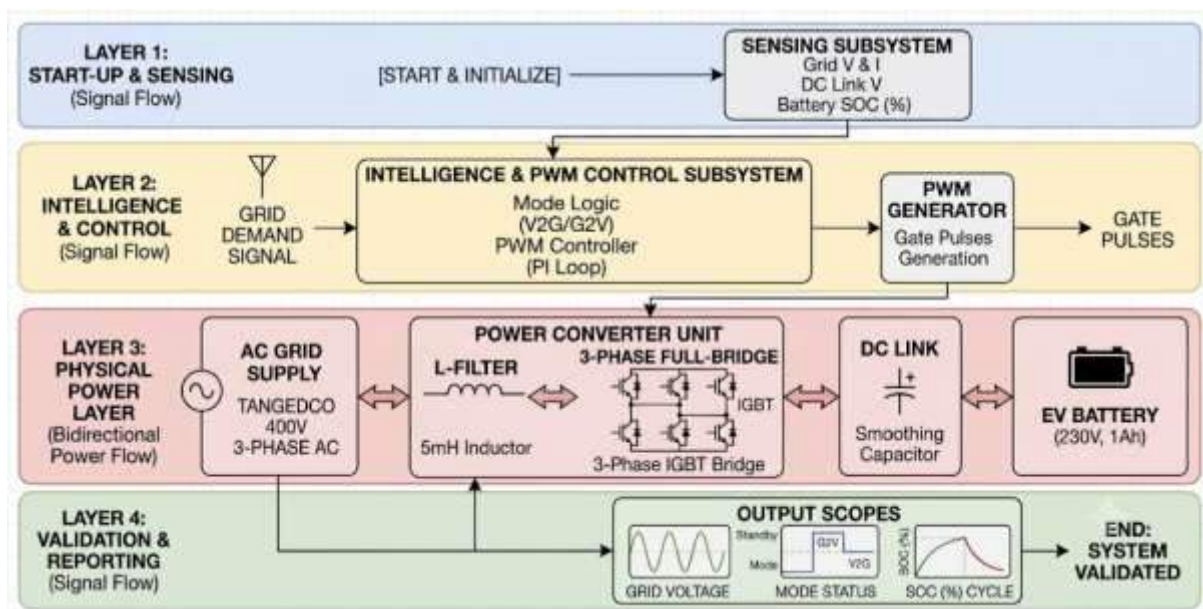


Figure 1: Block Diagram

## 2. Problem Statement

The rapid proliferation of Electric Vehicles (EVs) presents a dual-sided reality for modern power grids: it is both a significant challenge and an unprecedented opportunity. Currently, uncontrolled EV charging during peak demand periods places immense stress on grid infrastructure, leading to potential voltage instability and an increased reliance on fossil-fuel-powered peaker plants to meet the sudden surge in load.

Furthermore, the vast majority of existing EV charging architectures are strictly unidirectional. They treat the vehicle merely as a load that draws power from the grid, failing to harness the massive, combined energy storage potential of parked EVs during times of high grid stress.

This project addresses the critical need for a smart, bidirectional power interface between the EV battery ecosystem and the standard TANGEDCO 3-phase 400V/50Hz distribution grid. The proposed system must solve the following functional challenges:

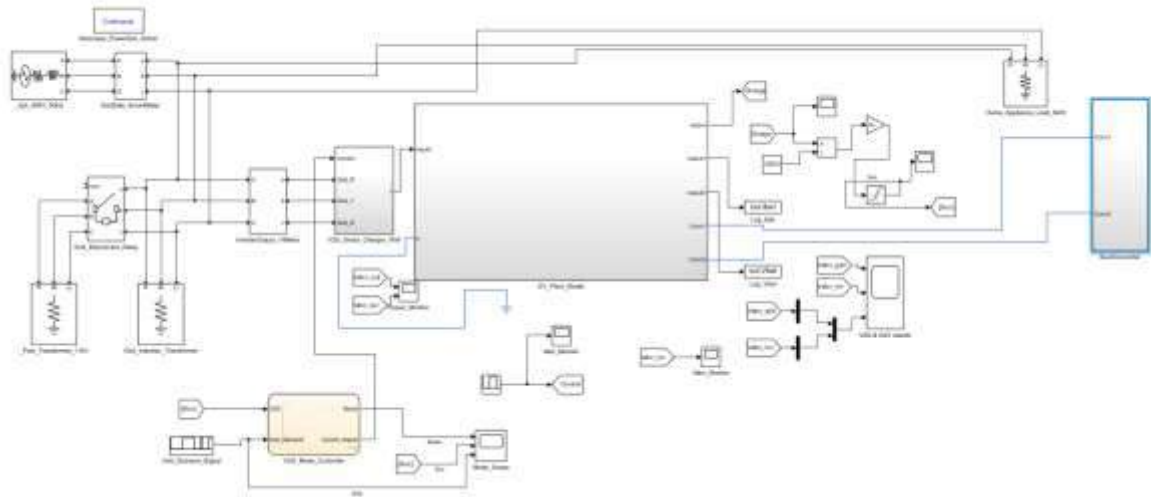
- **Intelligent Load Shifting:** Automatically draw power to charge the EV battery when overall grid demand is low (G2V mode).
- **Grid Support:** Feed stored, high-quality AC energy back into the local grid to alleviate stress during peak demand periods (V2G mode).
- **Asset Protection:** Actively protect the expensive EV battery assets from over-charging and deep-discharging limits.
- **Autonomous Operation:** Execute all routing and switching decisions autonomously utilizing a centralized, Stateflow-based intelligent logic controller.

## 3. System Architecture & Design Methodology

### 3.1 Top-Level Model Architecture

The top-level Simulink model represents the complete macro-system encompassing the grid, the conversion interface, and the vehicle. The grid is modeled as a 3-phase 400V, 50Hz AC source. Power flows through simulated distribution components, including a pole transformer and grid injection transformers.

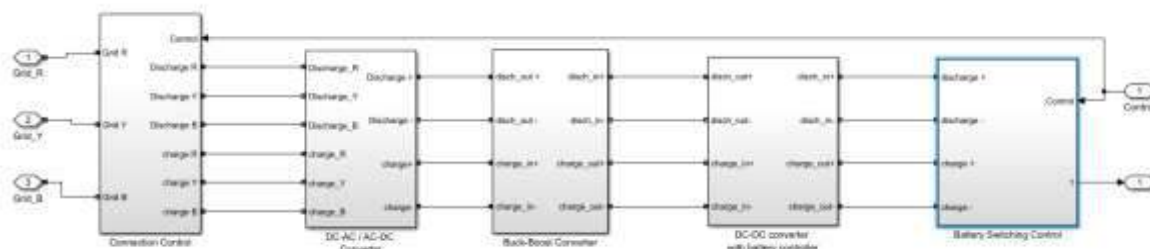
A `Grid_Disconnect_Relay` acts as a primary safety mechanism. Standard V-I measurement blocks continuously log voltage and current waveforms at the interface points. The power lines interface directly with the `V2G_Smart_Charger_7kW` block, which is bi-directionally connected to the `EV_Plant_Model`



**Fig. 2 Top-Level Simulink Model Architecture for Bidirectional V2G System**

### 3.2 V2G Smart Charger Subsystem

The Smart Charger subsystem is the physical power electronics layer responsible for bidirectional power conversion. To maintain clean sinusoidal AC waveforms and a highly regulated DC output, the subsystem is modularized into sequential conversion stages: connection control, an active front-end DC-AC/AC-DC converter using a 3-phase full-bridge IGBT configuration, and an internal L-filter (5mH) to smooth current ripples.



**Fig. 3 V2G Smart Charger Power Electronics Internal Subsystem**

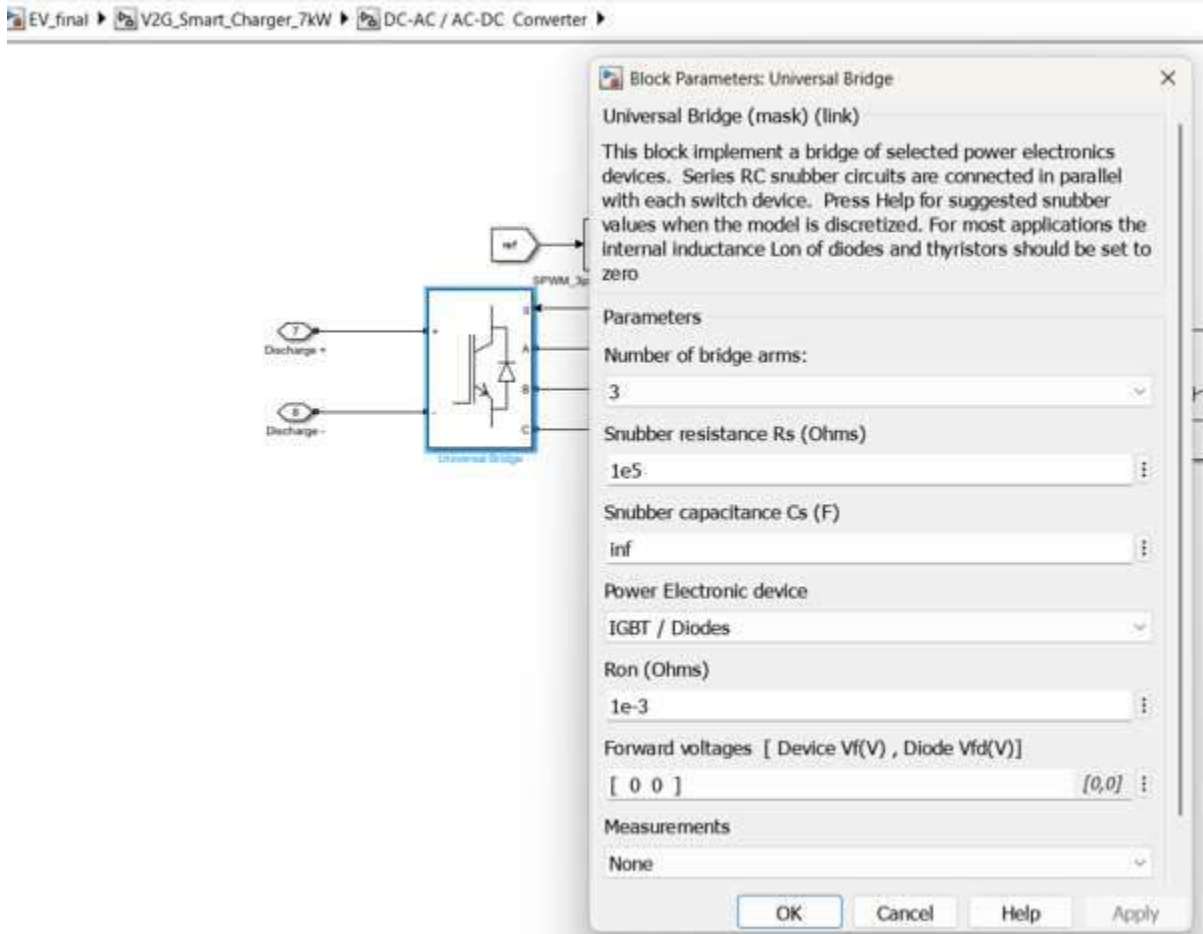
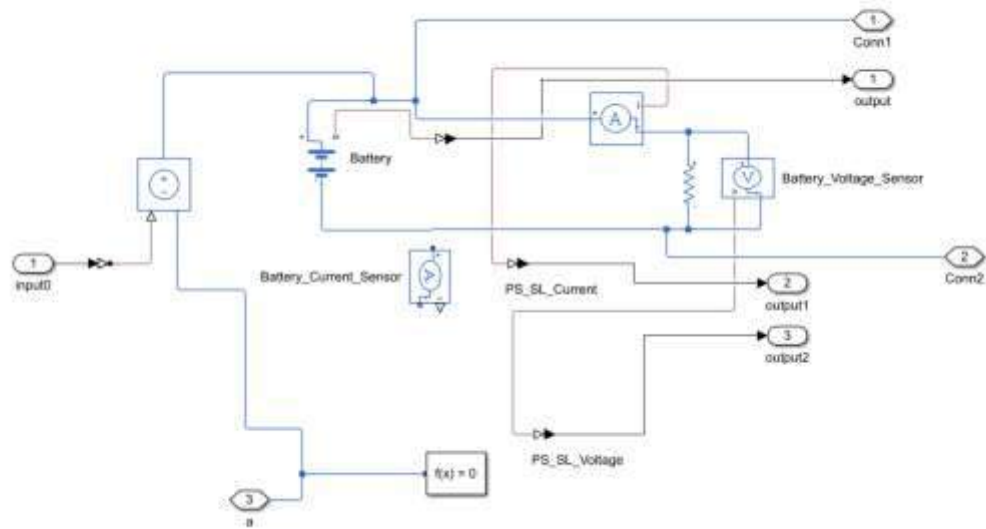


Fig .4 Inside V2G Subsystem DC to AC Converter IGBT

### 3.3 EV Plant & Battery Model

The EV\_Plant\_Model represents the physical characteristics of the Electric Vehicle's energy storage system. It features a battery block parameterized to 230V, 1Ah to capture transient dynamics and SOC depletion/replenishment curves. Current and voltage sensors act as the measurement tools to feed real-time battery status back to the controller.

A scaled-down capacity of 1Ah is purposefully utilized in the behavioral simulation layer to accelerate execution times and validate transient state transitions without waiting for extended physical thermal or chemical cycling."



**Fig. 5 Battery Model**



Block Parameters: Battery

Battery Auto Apply

| Settings                                       | Description | VALUE |  |
|------------------------------------------------|-------------|-------|--|
| <b>Main</b>                                    |             |       |  |
| Nominal voltage, Vnom                          | 230         | V     |  |
| Current directionality                         | Enabled     |       |  |
| Internal resistance during discharging         | 0.1         | Ohm   |  |
| Internal resistance during charging            | 0.1         | Ohm   |  |
| Battery charge capacity                        | Finite      |       |  |
| Cell capacity (Ah rating)                      | 1           | A*hr  |  |
| Voltage V1 when charge is AH1                  | 207         | V     |  |
| Charge AH1 when no-load voltage is V1          | 0.4         | A*hr  |  |
| Self-discharge                                 | Disabled    |       |  |
| Expose charge measurement port                 | Yes         |       |  |
| <b>Dynamics</b>                                |             |       |  |
| <b>Fade</b>                                    |             |       |  |
| <b>Calendar Aging</b>                          |             |       |  |
| <b>Thermal Port</b>                            |             |       |  |
| <b>Initial Targets</b>                         |             |       |  |
| <input type="checkbox"/> Current (positive in) |             |       |  |
| <input type="checkbox"/> Voltage               |             |       |  |
| <input checked="" type="checkbox"/> Charge     |             |       |  |
| Priority                                       | High        |       |  |
| Value                                          | 0.4         | A*hr  |  |
| <b>Nominal Values</b>                          |             |       |  |

Fig . 6 Battery Cell Parametrization Layer

### 3.4 Buck-Boost & DC-DC Converter Logic

A secondary DC-DC stage steps up or steps down the voltage to match the EV battery's strict voltage requirements. A dedicated buck converter subsystem containing a buck inductor, a capacitor, and an RL load is integrated to represent the internal auxiliary DC loads of the vehicle, ensuring the simulation accounts for parasitic drain.

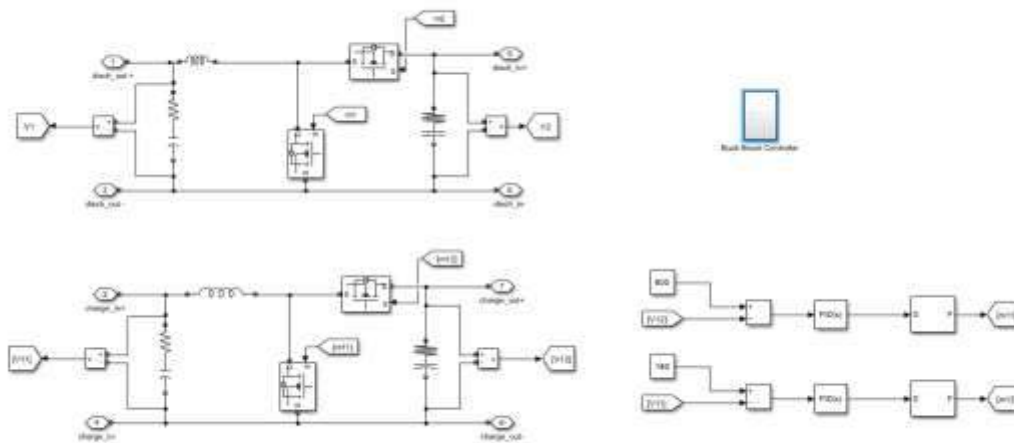


Fig. 7 Buck-Boost Converter and Auxiliary DC Vehicle Load Logic

### 3.5 Stateflow Controller (Mode Switching)

The operational brain of the V2G system is the V2G\_Mode\_Controller, implemented via Simulink Stateflow. It operates as a finite state machine managing transitions between:

- **STANDBY (Mode 0):** Idle safety state when SOC thresholds are reached ( $SOC \geq 60$  or  $SOC \leq 20$ ).
- **G2V\_CHARGING (Mode 1):** Active charging state triggered when  $Grid\_Demand == 0$  and  $SOC < 60$ .
- **V2G\_DISCHARGING (Mode 2):** Energy injection mode triggered during peak stress when  $Grid\_Demand == 1$  and  $SOC > 40$ .

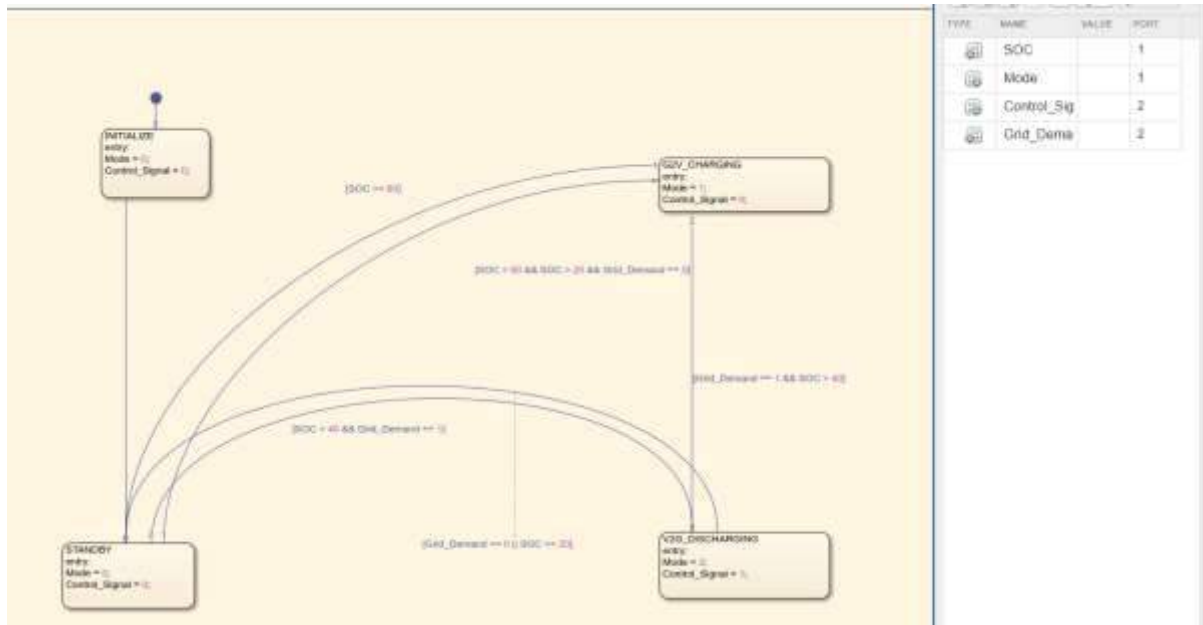


Fig . 8 Stateflow Finite State Machine Controller Logic

## 4. Simulation Setup & Solver Selection

Given the hybrid nature of this model—combining high-frequency discrete power electronics switching with continuous physical plant dynamics—the selection of the correct numerical solver is critical.

- **Solver Type:** Variable-step continuous solver.
- **Solver Selection:** ode23tb (stiff/TR-BDF2). This selection efficiently handles the mathematical stiffness introduced by the instantaneous switching of the IGBTs and diodes within the Smart Charger block without inducing numerical instability or forcing excessively small step sizes.
- **Control Sample Time:** The Stateflow controller executes at a discrete sample rate to mirror a realistic embedded microprocessor clock speed.

## 5. Simulation Results & Waveform Analysis

### 5.1 Stateflow Logic Validation (Mode Scope)

The Mode\_Scope output successfully verifies that the controller autonomously transitions between the operational states:

- $t = 0.0s$  to  $0.4s$ : Operates in Mode 1 (G2V Charging) while demand is 0, showing a linear increase in battery SOC.
- $t$  (approx.)  $0.42s$ : Automatically switches to Mode 2 (V2G Discharging) as the grid demand signal transitions to 1.
- $t$  (approx.)  $0.70s$ : Transitions smoothly to Mode 0 (Standby) when the battery hits its upper charging boundary, flattening the SOC trace to protect cell health.



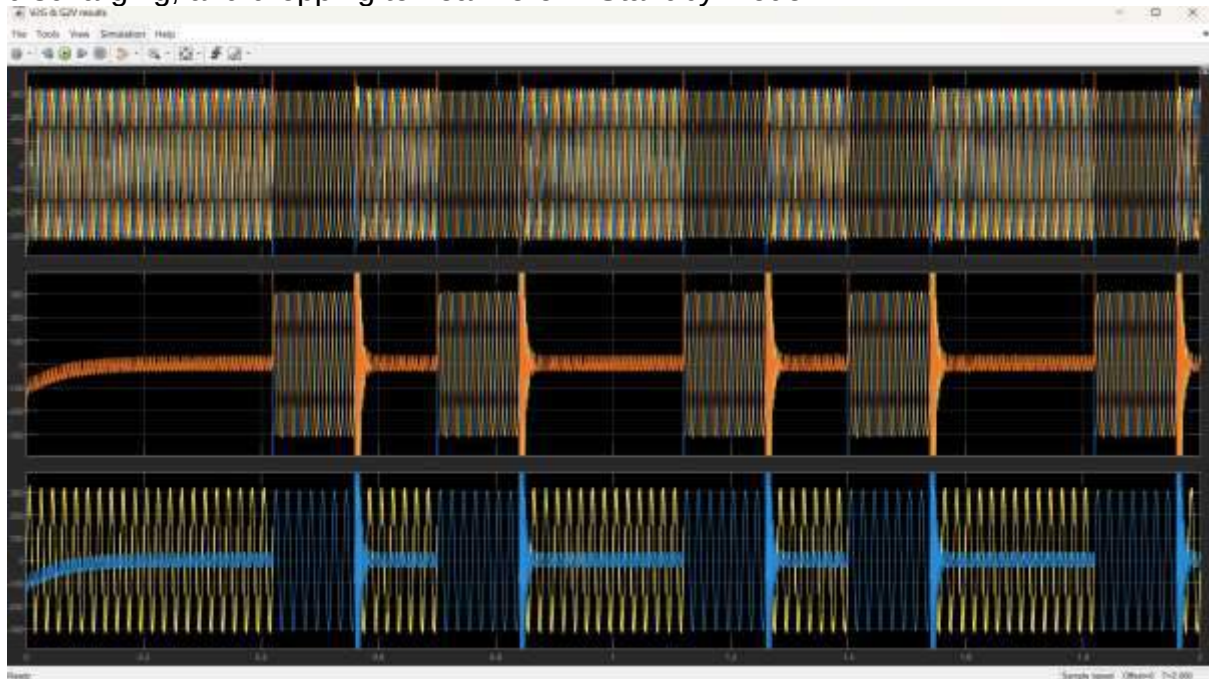
Fig. 9 Mode Scope Verifying Stateflow Transitions and SOC Profile

|    |     |   |               |                                         |
|----|-----|---|---------------|-----------------------------------------|
| 1  | 50% | 0 | G2V Charging  | SOC < 60 + no demand → charges          |
| 2  | 55% | 0 | G2V Charging  | Still below 60, stays charging          |
| 3  | 58% | 0 | G2V Charging  | Just under threshold                    |
| 4  | 60% | 0 | Standby       | SOC hits 60 → standby (no demand)       |
| 5  | 65% | 1 | V2G Discharge | SOC > 40 + demand=1 → discharges        |
| 6  | 55% | 1 | V2G Discharge | Still above 40, keeps discharging       |
| 7  | 42% | 1 | V2G Discharge | Just above low threshold                |
| 8  | 20% | 1 | Standby       | SOC hits 20 → standby (protect battery) |
| 9  | 25% | 0 | G2V Charging  | Demand off, SOC < 60 → charges again    |
| 10 | 70% | 0 | Standby       | SOC >= 70 → upper standby cutoff        |
| 11 | 72% | 1 | V2G Discharge | Demand arrives at high SOC → V2G        |
| 12 | 45% | 0 | G2V Charging  | Demand gone, SOC mid → charges          |
| 13 | 18% | 0 | Standby       | SOC <= 20 → lower cutoff standby        |
| 14 | 10% | 1 | Standby       | Even with demand, SOC too low           |

Fig. 10 Stateflow Logic

## 5.2 AC Grid Interface Waveforms

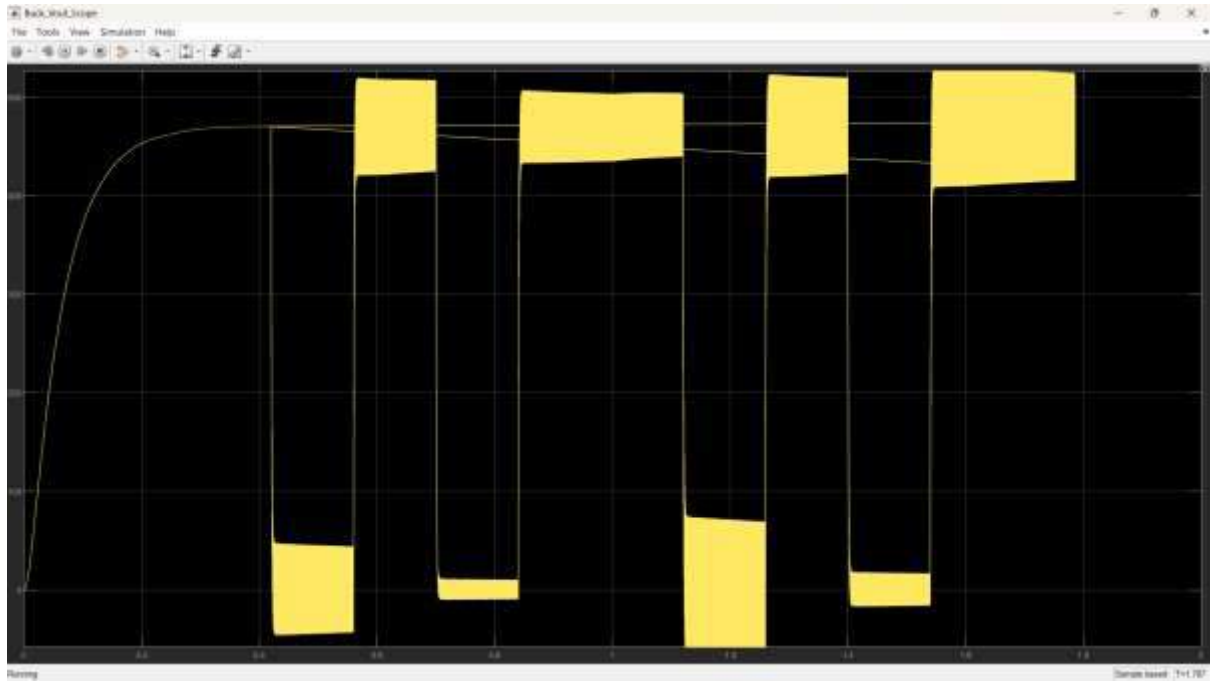
The three-phase AC scopes monitor the physical power quality at the point of common coupling. The grid voltage remains a stable, clean sinusoidal waveform throughout all mode transitions. The current waveform clearly illustrates bidirectional behavior: drawing current during G2V charging, reversing phase to inject current during V2G discharging, and dropping to near zero in Standby mode.



**Fig. 11 Three-Phase AC Grid Interface Voltage and Current Waveforms**

## 5.3 DC Link & Buck Converter Transients

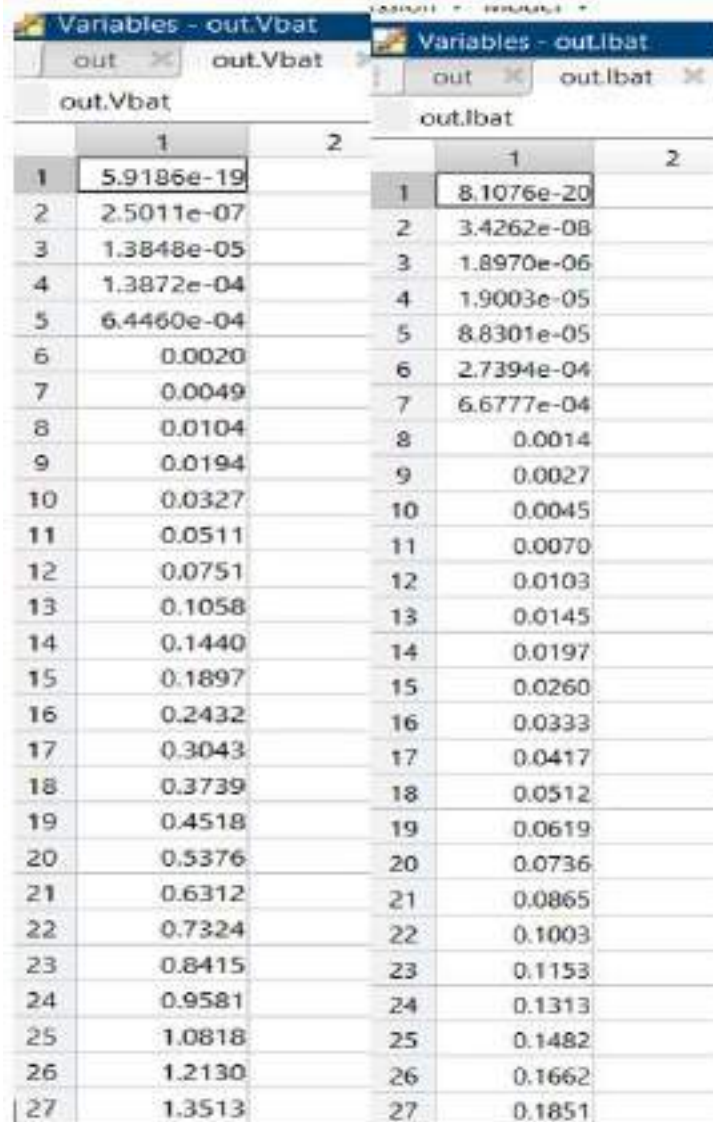
The output voltage profile of the auxiliary Buck Converter demonstrates steady-state regulation. It exhibits sharp transient voltage spikes and dips at the exact moments the system performs high-power mode switching, characterizing the inductive energy steps during rapid current reversals.



**Fig. 12 Transient Voltage Output Waveform of the DC Buck Converter**

## 5.4 Battery Scope Waveform Performance

**Dedicated workspace data arrays (out.Ibat and out.Vbat)** captured in the MATLAB Variable Editor monitor cell-level galvanic behavior, staying positive during G2V charging, matching zero during standby isolation, and turning negative during active V2G extraction. The voltage scope shows stable compliance around the nominal 230V pack parameters, maintaining minimal sag during high-power handoffs.



| Variables - out.Vbat |            | Variables - out.Ibat |            |
|----------------------|------------|----------------------|------------|
| out.Vbat             |            | out.Ibat             |            |
|                      | 1          |                      | 1          |
| 1                    | 5.9186e-19 | 1                    | 8.1076e-20 |
| 2                    | 2.5011e-07 | 2                    | 3.4262e-08 |
| 3                    | 1.3848e-05 | 3                    | 1.8970e-06 |
| 4                    | 1.3872e-04 | 4                    | 1.9003e-05 |
| 5                    | 6.4460e-04 | 5                    | 8.8301e-05 |
| 6                    | 0.0020     | 6                    | 2.7394e-04 |
| 7                    | 0.0049     | 7                    | 6.6777e-04 |
| 8                    | 0.0104     | 8                    | 0.0014     |
| 9                    | 0.0194     | 9                    | 0.0027     |
| 10                   | 0.0327     | 10                   | 0.0045     |
| 11                   | 0.0511     | 11                   | 0.0070     |
| 12                   | 0.0751     | 12                   | 0.0103     |
| 13                   | 0.1058     | 13                   | 0.0145     |
| 14                   | 0.1440     | 14                   | 0.0197     |
| 15                   | 0.1897     | 15                   | 0.0260     |
| 16                   | 0.2432     | 16                   | 0.0333     |
| 17                   | 0.3043     | 17                   | 0.0417     |
| 18                   | 0.3739     | 18                   | 0.0512     |
| 19                   | 0.4518     | 19                   | 0.0619     |
| 20                   | 0.5376     | 20                   | 0.0736     |
| 21                   | 0.6312     | 21                   | 0.0865     |
| 22                   | 0.7324     | 22                   | 0.1003     |
| 23                   | 0.8415     | 23                   | 0.1153     |
| 24                   | 0.9581     | 24                   | 0.1313     |
| 25                   | 1.0818     | 25                   | 0.1482     |
| 26                   | 1.2130     | 26                   | 0.1662     |
| 27                   | 1.3513     | 27                   | 0.1851     |

Fig . 13 The Workspace data for current and voltage



## 6. Model Advisor Report

The Model Advisor tool was executed on the EV\_final system to verify compliance with industry modeling guidelines and to ensure structural safety before generating deployable code.

### 6.1 Baseline Analysis Metrics

The Model Advisor checked the structural integrity of the blocks, uncovering zero errors or fatal defects. The remaining warnings were justified as expected behavior for non-runtime parameters.

- **Passed:** 165 checks
- **Warning:** 39 checks
- **Justified:** 1 check
- **Failed / Incomplete:** 0 checks
- **Total Checked Items:** 205

### 6.2 Upgrade Advisor Checks

An additional verification run via the Upgrade Advisor ensured compatibility with current Simulink version execution environments.

- **Passed:** 45 checks
- **Warning:** 2 checks (related to test harness upgrade sequence dependencies)
- **Failed:** 0 checks

## 7. Coverage Report

Model coverage was generated using Simulink Test Manager to evaluate the execution completeness of the testing vectors applied to the model.

### 7.1 Aggregated Model Coverage Metrics

The verification run labeled MIL\_FullModel simulated the system under a normal execution framework over a 2.0-second time stop. The execution achieved comprehensive verification across all core logic states:

| Coverage Metric    | Achieved Percentage |
|--------------------|---------------------|
| Execution Coverage | 99%                 |
| Condition Coverage | 86%                 |
| Decision Coverage  | 65%                 |
| MCDC Coverage      | 62%                 |
| Model Complexity   | 74%                 |
|                    |                     |



## 8. Simulink Coder & SIL Verification

### 8.1 Code Generation Parameters

Using Simulink Coder, the high-level V2G\_Controller\_SIL model was compiled into production-ready source code optimized for embedded environments.

- **System Target File:** ert.tlc (Embedded Coder Target)
- **Hardware Device Type:** Intel->x86-64 (Windows64)
- **Simulink Coder Version:** 24.2
- **Build Type:** Model reference compilation

### 8.2 Code Interface Summary

The generated code encapsulates the primary outputs within structured, accessible memory definitions. The code interface report maps the root-level model outputs to production structures:

- **V2G\_Controller\_SIL\_Y.Mode:** Output data structure for the active controller mode, typed as real\_T.
- **V2G\_Controller\_SIL\_Y.Control\_Signal:** Output data structure for the physical converter gate trigger commands, typed as real\_T.

```
/* Generated Outport Structure Excerpt */  
struct {  
    real_T Mode;  
    real_T Control_Signal;  
} V2G_Controller_SIL_Y;
```

### 8.3 Software-in-the-Loop (SIL) Testing

To validate the compiled code accuracy against the golden simulation, a Software-in-the-Loop test case named SIL\_Controller was initiated. The code compiled successfully and completed execution in 70.91 seconds under a FixedStepDiscrete solver configuration with a fixed step size of 0.01. The simulation returned a **Passed** status, proving that the generated C code output exactly matches the behavioral model logic.

## 9. Debugging Techniques

The development and integration of the V2G cyber-physical architecture leveraged systematic MBD debugging methods to isolate faults and evaluate signal flows:

- **Log Signals:** Used to tag and record the precise trace variables across simulation runs for post-execution examination within the Simulation Data Inspector.

- **Port Value Label:** Enabled on the active model canvas to display real-time signal values directly next to input/output block pins during simulation steps, accelerating logic routing checks.
- **Breakpoints:** Injected at specific conditional block boundaries to pause model execution precisely when a value matches user-defined criteria, facilitating isolation of unexpected transients.
- **Execution Order Visualization:** Applied to check the exact sorting order of concurrent subsystem blocks, ensuring control outputs update before the physical converter blocks process them, eliminating algebraic loops.

## 10. Student Reflections & Future Scope

### Student Reflections:

This project served as a profound learning experience, bridging the gap between theoretical power electronics and applied embedded systems logic. Developing this bidirectional Vehicle-to-Grid architecture utilizing Model-Based Design (MBD) fundamentally shifted our perspective on cyber-physical systems engineering. Rather than treating software and hardware as isolated components, we learned how to design a cohesive workflow where finite state machines (implemented in Stateflow) directly manage high-voltage physical switching elements (modeled in Simscape).

As a team, navigating the complexities of multi-domain simulations, resolving algebraic loop dependencies, and validating code generation through Software-in-the-Loop (SIL) testing sharpened our system debugging and data validation skill sets. It provided critical insight into how modern software-defined vehicles and grid-tied systems are engineered in real-world industry settings.

### Future Scope:

While the current simulation model successfully verifies the bidirectional energy transfer concept, several pathways exist to expand this research into production-ready deployment:

1. **Hardware-in-the-Loop (HIL) Testing:** Generate production C-code using Embedded Coder and deploy the controller onto physical microcontrollers (e.g., an ESP32 or STM32 platform) linked to hardware channel emulators.
2. **Advanced Control Implementations:** Substitute the baseline PID loop blocks with fuzzy logic or predictive current control strategies to mitigate the high transient voltage surges observed during micro-second switching intervals.
3. **Automotive Standard Conformance:** Adapt the generated code architecture to align with modern AUTOSAR runtime frameworks to facilitate integration into standard Electric Vehicle Electronic Control Units (ECUs).

## A Appendix A: MATLAB Onramp Certificates



### Course Completion Certificate

INDHU PRAKASH

has successfully completed **100%** of the self-paced training course

MATLAB Onramp

  
DIRECTOR, TRAINING SERVICES

6 April 2026

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Fig . 14 Certificate 1– MATLAB Onramp



### Course Completion Certificate

INDHU PRAKASH

has successfully completed **100%** of the self-paced training course

Simscape Onramp

  
DIRECTOR, TRAINING SERVICES

2 April 2026

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Fig . 15 Certifiacte 2 – SIMSCAPE Onramp



## Course Completion Certificate

INDHU PRAKASH

has successfully completed **100%** of the self-paced training course

Simulink Onramp



DIRECTOR, TRAINING SERVICES

1 December 2025

Fig. 16 Certificate 3 – SIMULINK Onramp



## Course Completion Certificate

INDHU PRAKASH

has successfully completed **100%** of the self-paced training course

Stateflow Onramp



DIRECTOR, TRAINING SERVICES

7 April 2026

Fig .17 Certificate 4 – STATEFLOW Onramp

## B Appendix B: Model Advisor Report

## Run Summary

**Incomplete Failed Warning Justified Passed Information Not Run Total**

 0
  0
  39
  1
  165
  0
  0
 205

Fig .18 Model Advisor Report

## C Appendix C: Coverage Report

---

**Test Case Information**

Name: MIL\_FullModel  
 Type: Simulation Test

**Aggregated Coverage Results**

| Analyzed Model                                                                                               | Sim Mode | Complexity | Decision | Condition | MCDC | Execution |
|--------------------------------------------------------------------------------------------------------------|----------|------------|----------|-----------|------|-----------|
|  <a href="#">EV_final</a> | Normal   | 74         | 65%      | 86%       | 62%  | 99%       |

Fig .19 Coverage Report

## D Appendix D: Code Generation Report

Target selection:

|                          |                |                                               |
|--------------------------|----------------|-----------------------------------------------|
| System target file:      | ert.tlc        | Browse...                                     |
| Description:             | Embedded Coder |                                               |
| Shared coder dictionary: | <empty>        | Set up...                                     |
| Language:                | C              |                                               |
| Language standard:       | C99 (ISO)      | Select code generation language               |
|                          |                | <input type="checkbox"/> Generate Halide code |

Build process:

☒ Generate code only  
☐ Package code and artifacts

Toolchain: Automatically locate an installed toolchain  
MinGW64 | gmake (64-bit Windows)

Build configuration: Faster Builds

► Toolchain details

Code generation objectives:

Prioritized objectives: Unspecified

Check model before generating code: Off

Set Objectives...  
Check Model...

**Fig. 20 Simulink Coder– Configuration Settings**

## 11. Mapping the project with POs, PSOs, and SDGs

Title of the Project : Model Based Design of Bidirectional Vehicle-to-Grid (V2G)

Power Conversion System

Project Member Name & Register Number : INDHU PRAKASH .S

(CB.EN.U4ELC23019)

### **PO mapping:**

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

### **Program Specific Outcomes (PSOs) :**

PSO1: Design and develop autonomous systems by integrating knowledge of electrical and computer engineering domains.

PSO2: Apply computational and networking skills for secured smart electrical systems.

PSO3: Innovate sustainable solutions for interdisciplinary domains using emerging technologies and data-driven approaches.



| SDGs                                                                                                                                  | Justification to the Contribution                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>7</b> AFFORDABLE AND CLEAN ENERGY<br>             | <p>The bidirectional system handles peak grid demand by injecting stored EV battery energy back into the network (V2G mode). This dynamic peak-shaving minimizes reliance on fossil-fuel peaker plants and improves grid reliability.</p> |
| <b>9</b> INDUSTRY, INNOVATION AND INFRASTRUCTURE<br> | <p>Utilizes advanced Model-Based Design (MBD) to simulate real-world cyber-physical automotive architectures. Proposes upgrading traditional infrastructure with software-defined vehicle integration and AUTOSAR compliance.</p>         |
| <b>11</b> SUSTAINABLE CITIES AND COMMUNITIES<br>    | <p>Solves the urban challenge of massive grid stress caused by uncontrolled, simultaneous EV charging. It establishes an intelligent load-shifting framework to create resilient, smart city microgrids</p>                               |
| <b>13</b> CLIMATE ACTION<br>                       | <p>By optimizing energy flow and reducing peak-load stress , the model lowers overall carbon emissions associated with emergency fossil-fuel power generation , accelerating the transition to clean transportation.</p>                  |